

ex:- $A = \{0, 2, 4, 6, 8\}$ $B = \{1, 3, 5, 7\}$

$U = \{0, 1, 2, \dots, 8\}$

$\bar{A} = ? U - A = \{1, 3, 5, 7\}$

$\bar{B} = ? U - B = \{0, 2, 4, 6, 8\}$

$A \cap \bar{A} = ? \emptyset$ (Rule 1)

$A \cup \bar{A} = ? U = \{0, 1, 2, 3, 4, 5, 6, 7, 8\} \Rightarrow U$ (R. 2)

$A - \bar{A} = ? A = \{0, 2, 4, 6, 8\}$ (R. 3)

$\overline{A \cup B} = ? A \cup B = U \Rightarrow \overline{A \cup B} = \emptyset$

$\bar{A} \times B = ? \{1, 3, 5, 7\} \times \{1, 3, 5, 7\} = \{(x, y) : x, y \in B\}$

① $X = [-1, 3] \times [0, 2]$

$\rightarrow \bar{X} = ?$

$\rightarrow \bar{X} \cap [-2, 4] \times [-1, 3] = ?$

